

Homework Assignment 1: Slab-Geometry Boltzmann Equation

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1 Introduction & Case's Method

This report presents the solutions to Homework Assignment 1, which focuses on the monoenergetic, steady-state Boltzmann transport equation in slab geometry for an infinite, homogeneous, isotropic medium with an isotropic plane delta-source at $x = 0$. The transport equation is written as:

$$\mu \frac{\partial \psi(x, \mu)}{\partial x} + \psi(x, \mu) = \frac{c}{2} \int_{-1}^1 \psi(x, \mu') d\mu' + \frac{1}{4\pi} \delta(x) \quad (1)$$

where $\psi(x, \mu)$ is the angular flux, $c \in [0, 1]$ is the scattering ratio, and μ is the cosine of the angle of the particle's velocity with the x -axis.

In 1953, Kenneth Case analyzed this equation by searching for separation-of-variables solutions of the form:

$$\psi_\nu(x, \mu) = e^{-x/\nu} \phi_\nu(\mu) \quad (2)$$

Substituting this ansatz into the homogeneous transport equation leads to the eigenvalue problem:

$$(\nu - \mu) \phi_\nu(\mu) = \frac{c\nu}{2} \int_{-1}^1 \phi_\nu(\mu') d\mu' \quad (3)$$

Normalizing the eigenfunctions such that $\int_{-1}^1 \phi_\nu(\mu') d\mu' = 1$, we obtain:

$$(\nu - \mu) \phi_\nu(\mu) = \frac{c\nu}{2} \quad (4)$$

Depending on the value of the eigenvalue ν , two cases arise:

1. **Discrete Spectrum** ($\nu \notin [-1, 1]$): Since $\nu - \mu \neq 0$ for all $\mu \in [-1, 1]$, we can divide directly by $\nu - \mu$, yielding the discrete eigenfunctions:

$$\phi_0^\pm(\mu) = \frac{c\nu_0}{2(\nu_0 \mp \mu)} \quad (5)$$

Integrating these over $[-1, 1]$ and applying the normalization condition yields the transcendental characteristic equation for the discrete eigenvalues $\pm\nu_0$:

$$1 = c\nu_0 \operatorname{arctanh}(1/\nu_0) \quad (6)$$

For $c \in (0, 1)$, there is exactly one pair of real, positive roots $\pm\nu_0$ with $\nu_0 > 1$.

2. **Continuous Spectrum** ($\nu \in [-1, 1]$): In this case, the term $\nu - \mu$ vanishes at $\mu = \nu$. Thus, we cannot divide by $\nu - \mu$ in the sense of ordinary functions. Instead, in the theory of distributions (generalized functions), the general solution to the equation is:

$$\phi_\nu(\mu) = \frac{c\nu}{2} \mathcal{P} \frac{1}{\nu - \mu} + \lambda(\nu) \delta(\mu - \nu) \quad (7)$$

where $\mathcal{P}$ represents the Cauchy principal value distribution, $\delta(\mu - \nu)$ is the Dirac delta function, and $\lambda(\nu)$ is the dispersion function. Integrating $\phi_\nu(\mu)$ over $\mu \in [-1, 1]$ and setting the integral to 1 determines $\lambda(\nu)$:

$$\lambda(\nu) = 1 - c\nu \operatorname{arctanh}(\nu) \quad (8)$$

1.1 Intuition Behind the Continuous Spectrum and Generalized Functions

The presence of the Dirac delta function $\delta(\mu - \nu)$ and the principal value term $\mathcal{P} \frac{1}{\nu - \mu}$ in the continuous eigenfunctions $\phi_\nu(\mu)$ can feel counter-intuitive.

Physically, the continuous spectrum represents the singular behavior of particles that travel without undergoing collisions (uncollided transport) or undergo highly directional scattering. A particle traveling at a specific direction $\mu = \nu$ represents a singular distribution in angle, which is modeled by the Dirac delta $\delta(\mu - \nu)$. The principal value distribution $\mathcal{P} \frac{1}{\nu - \mu}$ accounts for the strong coupling between nearby angles due to scattering, where the singularity is balanced symmetrically.

However, when we look at the scalar flux $\phi(x)$, which is the angle-integrated quantity:

$$\phi(x) = 2\pi \int_{-1}^1 \psi(x, \mu) d\mu \quad (9)$$

these angular singularities are integrated out. Under integration, the distribution $\phi_\nu(\mu)$ acts as a linear functional mapping to a constant:

$$\int_{-1}^1 \phi_\nu(\mu) d\mu = 1 \quad (10)$$

This leads to the transient component of the scalar flux:

$$\phi_{\text{tr}}(x) = \frac{1}{2} \int_0^1 \frac{e^{-|x|/\nu}}{N_\nu} d\nu \quad (11)$$

where:

$$N_\nu = \nu \left[\lambda^2(\nu) + \frac{\pi^2 c^2 \nu^2}{4} \right] \quad (12)$$

Crucially, **no distributions or generalized functions remain in the scalar flux integral**. The integrand is a regular, smooth function of ν on the open interval $(0, 1)$. For any $x > 0$, as $\nu \rightarrow 0$, the exponential term $e^{-|x|/\nu}$ decays extremely rapidly, making the integrand vanish at 0. As $\nu \rightarrow 1$, the dispersion function $\lambda(\nu) \rightarrow -\infty$, which causes the normalization factor $N_\nu \rightarrow \infty$ and the integrand to vanish at 1. Thus, the numerical integration is highly well-behaved.

2 Question 1: Exact Scalar Flux Analysis

The exact scalar flux is computed as:

$$\phi(x) = \phi_{\text{as}}(x) + \phi_{\text{tr}}(x) \quad (13)$$

where $\phi_{\text{as}}(x) = \frac{e^{-|x|/\nu_0}}{2N_0^+}$ is the asymptotic component, and $\phi_{\text{tr}}(x)$ is the transient component. The discrete normalization factor is:

$$N_0^+ = \frac{c}{2}\nu_0^3 \left[\frac{c}{\nu_0^2 - 1} - \frac{1}{\nu_0^2} \right] \quad (14)$$

For ν_0 , we can choose to compute it using:

- **Numerical Root-Finding:** Solving $\frac{\text{arctanh}(y)}{y} = \frac{1}{c}$ for $y = 1/\nu_0 \in (0, 1)$ using a bisection method, which is extremely stable.
- **Analytic Approximation:**

$$\nu_0 \approx \frac{1}{\sqrt{1 - c^{p(c)}}} \quad (15)$$

where:

$$p(c) = 2.47412 + \frac{0.00363081}{c^2} - 0.0352458c + \frac{0.557498}{c} \quad (16)$$

2.1 Question 1a: Asymptotic and Transient Components

[Insert plots of asymptotic and transient components for each c , and the relative contributions $\phi_{\text{tr}}(x)/\phi(x)$ and $\phi_{\text{as}}(x)/\phi(x)$ as a function of x . Add physical discussion here.]

2.2 Question 1b: Classical Diffusion Solution

The classical diffusion equation (in units of mean free path, where $\Sigma_t = 1$) for a homogeneous, isotropic medium with an isotropic plane delta-source is given by:

$$-D \frac{d^2 \phi_{\text{class}}(x)}{dx^2} + (1 - c)\phi_{\text{class}}(x) = \delta(x) \quad (17)$$

Using the standard P1 diffusion coefficient $D = 1/3$, the Green's function solution is:

$$\phi_{\text{class}}(x) = \frac{\sqrt{3}}{2\sqrt{1-c}} e^{-\sqrt{3(1-c)}|x|} \quad (18)$$

[Insert comparison plots with the classical diffusion solution. Discuss the differences, including the behavior near the source ($x \rightarrow 0$) and the asymptotic behavior ($x \rightarrow \infty$).]

2.3 Question 1c: Asymptotic Diffusion Solution

The asymptotic diffusion approximation matches the diffusion length to the exact discrete transport eigenvalue ν_0 , i.e., $L = \nu_0$, which corresponds to an asymptotic diffusion coefficient $D_{\text{as}} = (1 - c)\nu_0^2$. The governing equation is:

$$-D_{\text{as}} \frac{d^2 \phi_{\text{as,diff}}(x)}{dx^2} + (1 - c)\phi_{\text{as,diff}}(x) = \delta(x) \quad (19)$$

which yields the Green's function solution:

$$\phi_{\text{as,diff}}(x) = \frac{1}{2(1-c)\nu_0} e^{-|x|/\nu_0} \quad (20)$$

[Insert comparison plots with the asymptotic diffusion solution. Discuss how the asymptotic diffusion solution compares to both the classical diffusion solution and the exact transport asymptotic component $\phi_{\text{as}}(x)$.]

2.4 Question 1d: Relative Error of Diffusion Approximations

The relative error of each diffusion approximation relative to the exact transport solution is defined as:

$$\text{Relative Error (\%)} = \frac{|\phi_{\text{approx}}(x) - \phi_{\text{exact}}(x)|}{\phi_{\text{exact}}(x)} \times 100\% \quad (21)$$

[Insert relative error plots as a function of x . Add physical discussion comparing the accuracy of the classical and asymptotic diffusion approximations across the different values of c . Analyze why the error behaves as it does near the source and far from it.]

3 Question 2: Numerical Code for Diffusion Equation

To solve the time-independent diffusion equation with a delta-source at $x = 0$ numerically, we exploit the symmetry of the medium and integrate over the half-domain $x \in [-a, 0]$. Since the delta-source is located at $x = 0$, the domain $x < 0$ is source-free, governed by the homogeneous equation:

$$-D \frac{d^2 \phi(x)}{dx^2} + (1-c)\phi(x) = 0 \quad (22)$$

3.1 Analytical Conversion to a First-Order System

We convert this second-order ODE to a linear system of two first-order ODEs. Let:

$$y_1(x) = \phi(x), \quad y_2(x) = \frac{d\phi(x)}{dx} \quad (23)$$

This yields the first-order system:

$$\frac{d}{dx} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} y_2 \\ \frac{1-c}{D} y_1 \end{pmatrix} \quad (24)$$

3.2 Boundary Conditions and the Shooting Method

Because the exact solution decays exponentially as $e^{-\kappa|x|}$ where $\kappa = \sqrt{(1-c)/D}$, we choose the truncation boundary $x = -a$ where $a = 3/\kappa$ (corresponding to 3 decay lengths, where the flux is negligible).

We set the function boundary condition at the far boundary to zero:

$$y_1(-a) = 0 \quad (25)$$

and parameterize the unknown initial derivative as a shooting parameter s :

$$y_2(-a) = s \quad (26)$$

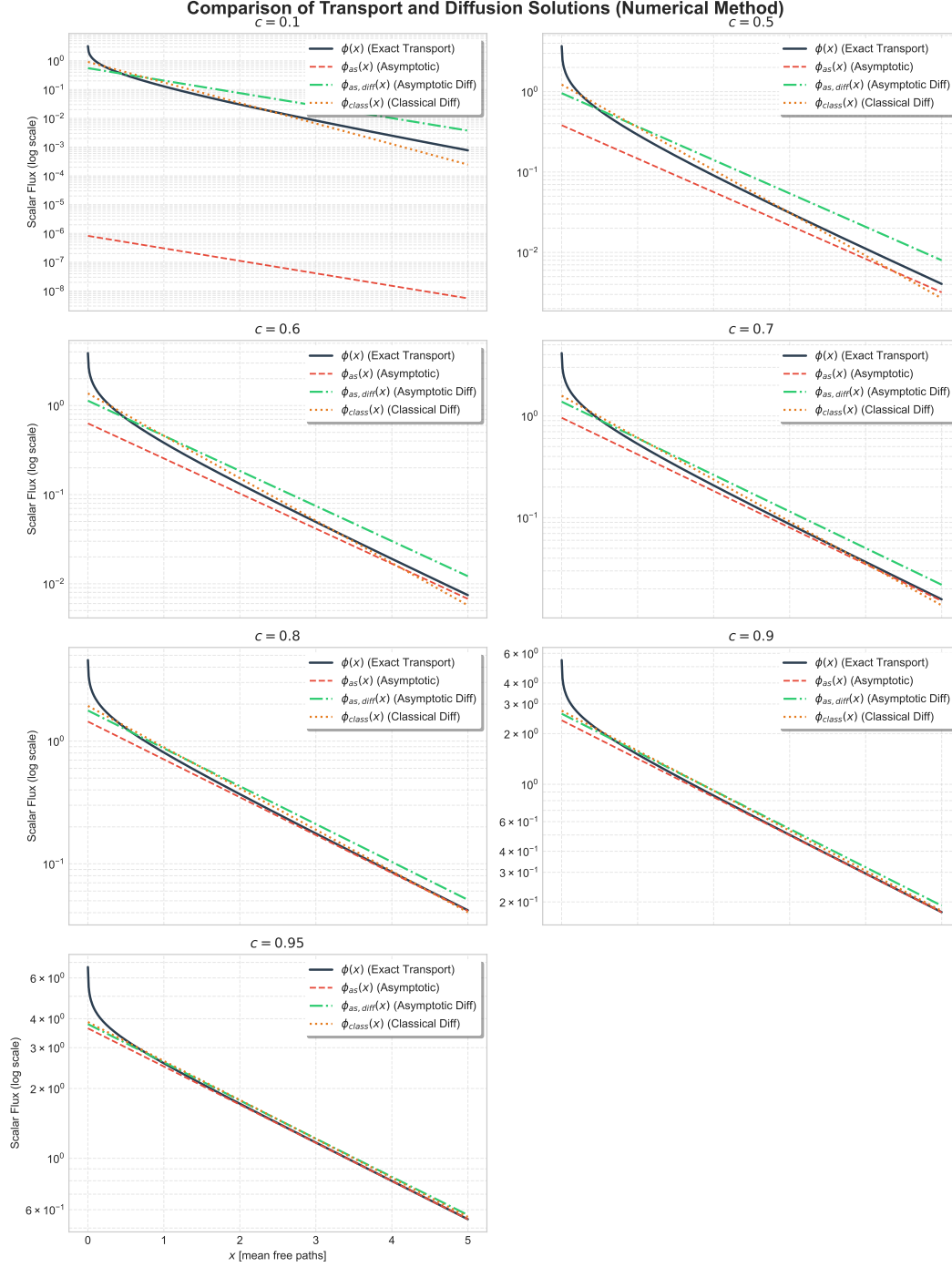


Figure 1: Comparison of the exact scalar flux, its asymptotic component, classical diffusion, and asymptotic diffusion solutions ($c = 0.5, 0.6, 0.7, 0.8, 0.9, 0.95$).

To find the correct s , we use the derivative jump condition at the source. Integrating the governing equation $-D\phi''(x) + (1 - c)\phi(x) = \delta(x)$ from $-\epsilon$ to $+\epsilon$ and taking the limit $\epsilon \rightarrow 0$ yields:

$$-D [\phi'(0^+) - \phi'(0^-)] = 1 \quad (27)$$

By symmetry of the infinite-medium solution ($\phi(x) = \phi(-x) \implies \phi'(0^-) = -\phi'(0^+)$), this simplifies

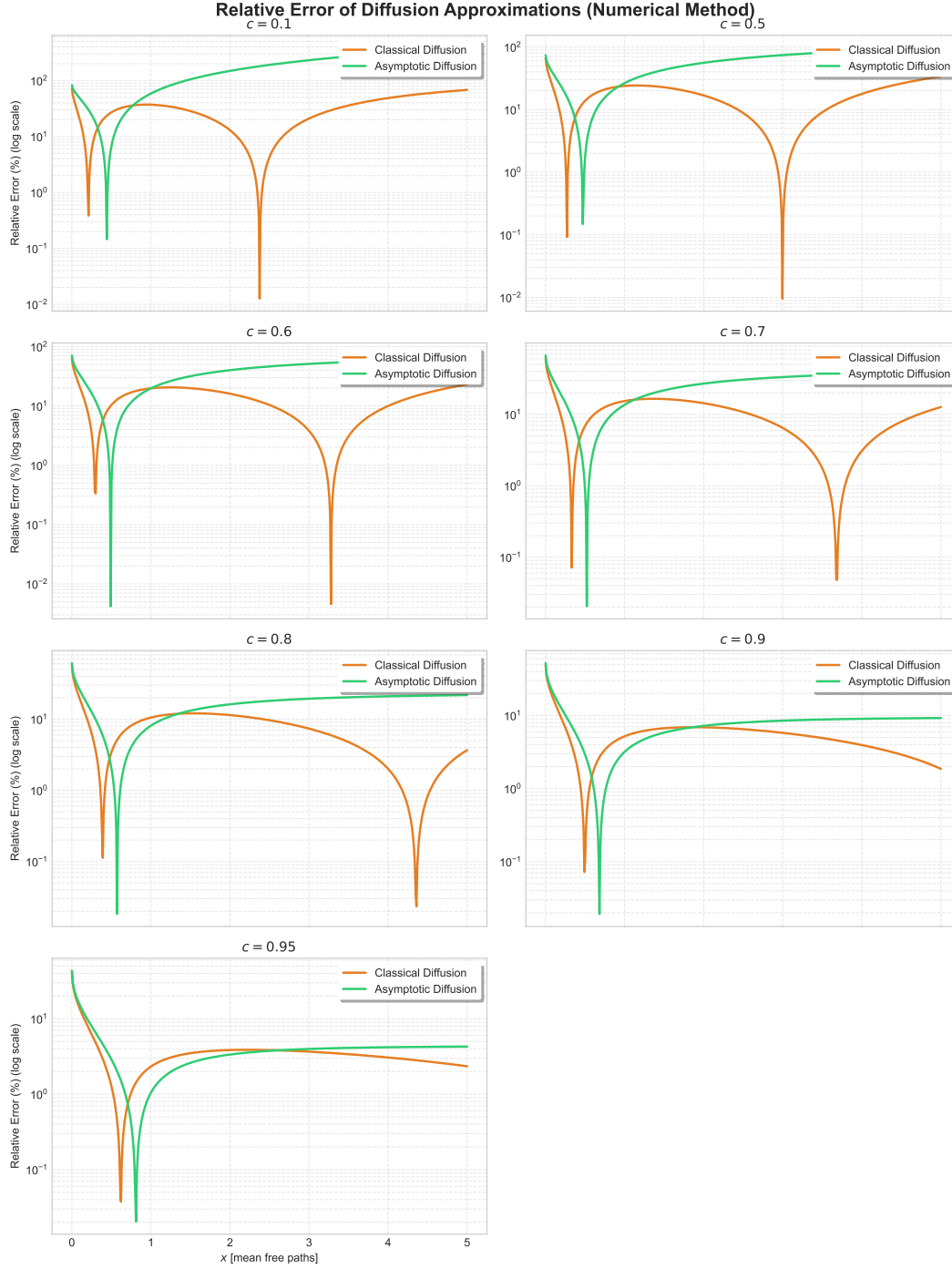


Figure 2: Relative error (%) of the classical and asymptotic diffusion approximations compared to the exact transport solution.

to the target boundary condition at $x = 0$:

$$y_2(0^-) = \phi'(0^-) = \frac{1}{2D} \quad (28)$$

Thus, we define the objective mismatch function:

$$f(s) = y_2(0; s) - \frac{1}{2D} \quad (29)$$

where $y_2(0; s)$ is obtained by integrating the first-order system from $-a$ to 0 with the initial state $[0.0, s]^T$. We use a bisection-based root-finding algorithm (specifically Brent's method, 'brentq') to solve $f(s^*) = 0$ for the optimal initial derivative s^* .

3.3 Comparison with the Analytical Solution

The numerical solution is compared to the analytical classical diffusion solution:

$$\phi_{\text{class}}(x) = \frac{1}{2\sqrt{D(1-c)}} e^{\kappa x} \quad (x < 0) \quad (30)$$

for three values of the scattering ratio: $c = 0.5, 0.7, 0.9$. The comparison of the flux values and the relative error are presented in Figure 3 and Figure 4.

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4 Question 3: Critical Half-Thickness and Critical Radius

[Present critical half-thickness $a/2$ and critical radius R_c results for $1.02 < c < 2$. Compare the various approximations.]

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5 Question 4: Numerical Diffusion Code in Spherical Coordinates

[Present the numerical results for critical radius calculations in spherical geometry, and compare with the analytical results.]

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6 Question 5: Bare Critical Mass of U-235 and Pu-239

[Present bare critical mass calculations using realistic cross sections.]

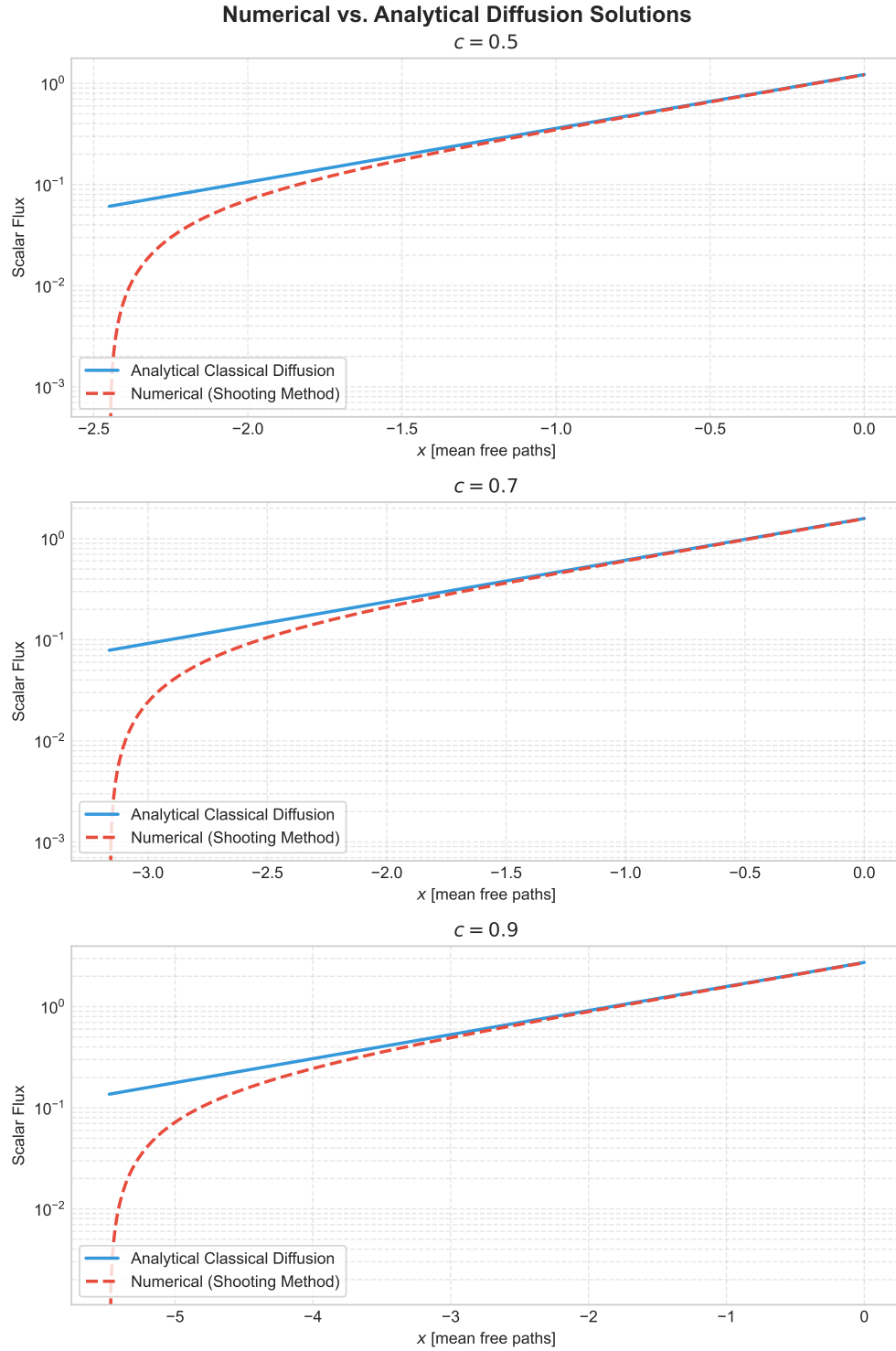


Figure 3: Comparison of the numerical (shooting method) solution and the analytical classical diffusion solution for $c = 0.5, 0.7, 0.9$ on the half-domain $[-a, 0]$.

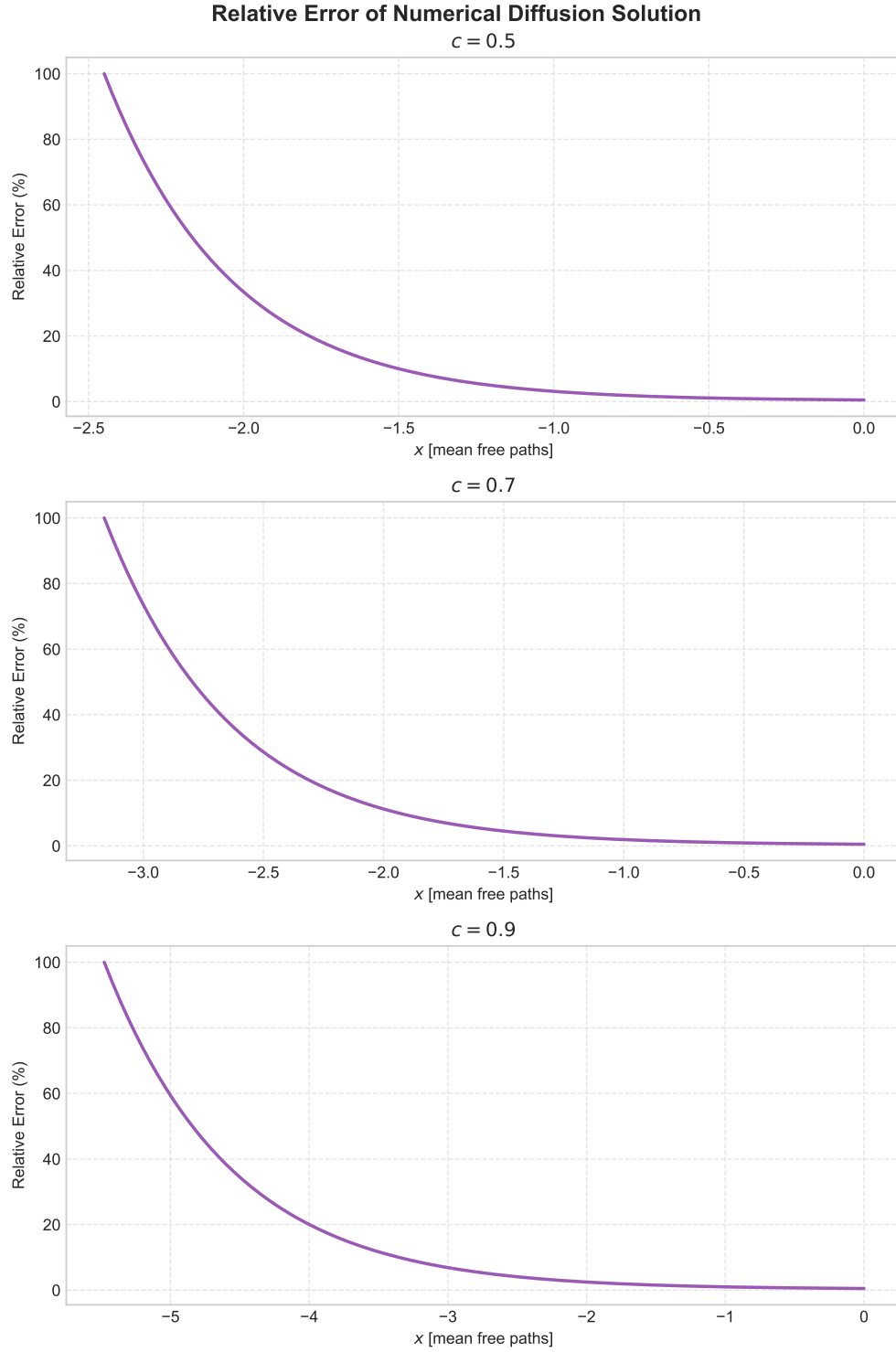


Figure 4: Relative error (%) of the numerical diffusion solution compared to the analytical classical diffusion solution on $[-a, 0]$.